

```
CREATE TABLE author (  
    idauthor INTEGER NOT NULL,  
    name    VARCHAR2(100) NOT NULL,  
    firstname VARCHAR2(100) NOT NULL  
);
```

```
ALTER TABLE author ADD CONSTRAINT author_pk PRIMARY KEY ( idauthor );
```

```
CREATE TABLE book (  
    idbook      INTEGER NOT NULL,  
    isbn        VARCHAR2(17) NOT NULL,  
    title       VARCHAR2(100) NOT NULL,  
    description  CLOB NOT NULL,  
    cover       VARCHAR2(100) NOT NULL,  
    availability CHAR(1) NOT NULL,  
    scoreavg    FLOAT,  
    author_idauthor INTEGER NOT NULL,  
    category_idcategory INTEGER NOT NULL  
);
```

```
ALTER TABLE book ADD CONSTRAINT book_pk PRIMARY KEY ( idbook );
```

```
ALTER TABLE book ADD CONSTRAINT book_isbn_un UNIQUE ( isbn );
```

```
CREATE TABLE category (  
    idcategory INTEGER NOT NULL,  
    name    VARCHAR2(100) NOT NULL  
);
```

```
ALTER TABLE category ADD CONSTRAINT category_pk PRIMARY KEY ( idcategory );
```

```
CREATE TABLE copy (  
    idcopy    INTEGER NOT NULL,  
    book_idbook INTEGER NOT NULL  
);
```

```
ALTER TABLE copy ADD CONSTRAINT copy_pk PRIMARY KEY ( idcopy );
```

```
CREATE TABLE loan (  
    idloan    INTEGER NOT NULL,  
    dateloan  TIMESTAMP NOT NULL,  
    datereturn TIMESTAMP,  
    copy_idcopy INTEGER NOT NULL,  
    user_iduser INTEGER NOT NULL  
);
```

```
ALTER TABLE loan ADD CONSTRAINT loan_pk PRIMARY KEY ( idloan );
```

```
CREATE TABLE member (  
    idmember INTEGER NOT NULL,  
    email    VARCHAR2(100) NOT NULL  
);
```

```
ALTER TABLE member ADD CONSTRAINT member_pk PRIMARY KEY ( idmember );
```

```
ALTER TABLE member ADD CONSTRAINT member_email_un UNIQUE ( email );
```

```
CREATE TABLE opinion (  
    idopinion INTEGER NOT NULL,  
    score    FLOAT NOT NULL,  
    "comment" CLOB,  
    book_idbook INTEGER NOT NULL,  
    user_iduser INTEGER NOT NULL  
);
```

```
ALTER TABLE opinion ADD CONSTRAINT opinion_pk PRIMARY KEY ( idopinion );
```

```
CREATE TABLE reservation (  
    idreservation INTEGER NOT NULL,  
    rank    INTEGER NOT NULL,  
    status   VARCHAR2(50) NOT NULL,  
    book_idbook INTEGER NOT NULL,  
    user_iduser INTEGER NOT NULL  
);
```

```
ALTER TABLE reservation ADD CONSTRAINT reservation_pk PRIMARY KEY ( idreservation  
);
```

```
CREATE TABLE "USER" (  
    iduser INTEGER NOT NULL,  
    email   VARCHAR2(100) NOT NULL,  
    password VARCHAR2(100) NOT NULL,  
    name    VARCHAR2(100) NOT NULL,  
    firstname VARCHAR2(100) NOT NULL,
```

```
phone VARCHAR2(10),  
role VARCHAR2  
-- ERROR: VARCHAR2 size not specified  
NOT NULL  
);
```

```
ALTER TABLE "USER" ADD CONSTRAINT user_pk PRIMARY KEY ( iduser );
```

```
ALTER TABLE "USER" ADD CONSTRAINT user_email_un UNIQUE ( email );
```

```
ALTER TABLE book  
ADD CONSTRAINT book_author_fk FOREIGN KEY ( author_idauthor )  
REFERENCES author ( idauthor );
```

```
ALTER TABLE book  
ADD CONSTRAINT book_category_fk FOREIGN KEY ( category_idcategory )  
REFERENCES category ( idcategory );
```

```
ALTER TABLE copy  
ADD CONSTRAINT copy_book_fk FOREIGN KEY ( book_idbook )  
REFERENCES book ( idbook );
```

```
ALTER TABLE loan  
ADD CONSTRAINT loan_copy_fk FOREIGN KEY ( copy_idcopy )  
REFERENCES copy ( idcopy );
```

```
ALTER TABLE loan  
ADD CONSTRAINT loan_user_fk FOREIGN KEY ( user_iduser )
```

```
REFERENCES "USER" ( iduser );
```

```
ALTER TABLE opinion
```

```
ADD CONSTRAINT opinion_book_fk FOREIGN KEY ( book_idbook )
```

```
REFERENCES book ( idbook );
```

```
ALTER TABLE opinion
```

```
ADD CONSTRAINT opinion_user_fk FOREIGN KEY ( user_iduser )
```

```
REFERENCES "USER" ( iduser );
```

```
ALTER TABLE reservation
```

```
ADD CONSTRAINT reservation_book_fk FOREIGN KEY ( book_idbook )
```

```
REFERENCES book ( idbook );
```

```
ALTER TABLE reservation
```

```
ADD CONSTRAINT reservation_user_fk FOREIGN KEY ( user_iduser )
```

```
REFERENCES "USER" ( iduser );
```